

Owen Garnier

Post-doctoral researcher in
Mathematics

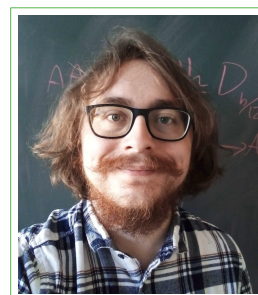
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Studies

- 2024 **PhD in mathematics**, LAMFA, Amiens.
Titre : Garside groupoids and complex braid groups.
Director : Ivan Marin
Other members of the Jury : Yves Guiraud, Jon McCammond, Adrien Brochier, Cédric Bonnafé, David Chataur, François Digne, Eddy Godelle.
- 2018–2021 **Masters degree in mathematics (Algebra, number theory and applications)**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).
Thesis: Théorie de Garside, applications aux groupes de tresses complexes. Cas de G_{31} . (supervisor: Ivan Marin).
- 2020 **Agrégation externe de mathématiques**, Rank: 10/387.
- 2015–2018 **Licence in mathematics**, Université de Picardie Jules Verne, Amiens.
With honors (Mention très bien).

Professional experiences

- 2025–2026 **Post-doctoral researcher**, University of Sevilla, supervised by María Cumplido Cabello.
- 2024–2025 **Temporary research and teaching assistant (ATER)**, Université de Picardie Jules Verne, Amiens.

Distinctions and prizes

- 2025 **PhD thesis award of the Université de Picardie Jules Verne**, Award given to the two best PhD thesis in science, technology and health subjects defended in 2024

Research papers

- 2024 **Uniqueness of roots up to conjugacy in circular and hosohedral-type Garside groups.**
Journal of Group Theory 27, 5 (2024), pp. 1091–1128.
- 2024 **Generalization of the Dehornoy-Lafont order complex to categories. Application to exceptional braid groups.**
Applied Categorical Structures 32, 1 (2024) Paper No. 1, 28 pp.
- 2023 **Regular theory in complex braid groups.**
Journal of Algebra 620 (2023), pp. 534–557.

Preprints

- 2025 **Generalized J -groups, J -braid groups and Seifert link groups**, Avec I. Haladjian

- 2025 **Conjugacy invariants and rigidity in Garside groups: a uniformity phenomenon**, *With M. Calvez, J. González-Meneses and B. Wiest*
- 2025 **Normalisers of parabolic subgroups of Artin-Tits groups and Tits cone intersections**, *With E. Heng, A. Licata and O. Yacobi*
- 2025 **Proof of Shvartsman's conjecture on braid groups of projective complex reflection groups.**
- 2024 **Parabolic subgroups of complex braid groups: the remaining case.**
- 2023 **Springer categories for regular centralizers in well-generated complex braid groups.**

Communications and Talks

Talks in conferences

- 2025 **Sous-groupes paraboliques des trickle groups**, *Workshop on trickle groups and their applications*, Tours.
Système de réécritures et solutions au problème du mot, *Workshop on trickle groups and their applications*, Tours.
Sous-groupes paraboliques des groupes de tresses complexes, *Journées jeunes chercheurs et chercheuses en topologie*, Amiens.
Groupeïdes de Garside et groupes de tresses complexes, *Colloque tournant du RT algèbre*, Clermont-Ferrand.
- 2023 **Homology computations for complex braid groups with the Dehornoy-Lafont order complex**, *2023 Meeting of the GDR Homotopy theory and applications*, Lille.
Bessis braid category for the complex braid group B_{31} , *Winterbraids XII*, Tours.
- 2022 **Regular theory in complex braid groups**, *Journées tresses 2022*, Amiens.

Talks in seminars

- 2025 **Normalisateurs des sous-groupes paraboliques dans les groupes d'Artin et de Coxeter**, *Séminaire de topologie de l'IMJ-PRG*, Paris
- 2025 **Normalizers of parabolic subgroups in Artin and Coxeter groups**, *Group theory seminar of the ICMAT*, Madrid (Spain)
Projective complex braid groups, *Algebra and Topology Seminar*, Canberra (Australia).
Sous-groupes paraboliques, des tresses aux groupeïdes de Garside, *Séminaire Géométrie, Algèbre, Dynamique et Topologie*, Dijon
Groupeïdes de Garside paraboliques et applications aux groupes de tresses complexes, *Séminaire d'algèbre et de géométrie*, Caen
- 2024 **Des partitions non-croisées à l'étude des sous-groupes paraboliques des groupes de tresses complexes**, *Séminaire SPACE*, Tours
Sous-groupes paraboliques et éléments réguliers des groupes de tresses complexes, *Gaston Darboux seminar of the IMAG*, Montpellier
Homology of categories and the Dehornoy-Lafont order complex, *Seminar of the "Ouragan" team*, Paris
- 2024 **Description de l'espace des orbites régulières d'un groupe de réflexions complexes**, *Topology seminar of the laboratoire Paul Painlevé*, Lille
- 2023 **Étude Garside du groupe de tresses complexe $B(G_{31})$** , *GATo Seminar of the LAMFA*, Amiens

Garside study of the complex braid group $B(G_{31})$, Seminar “Groupes, Représentations et Géométrie” of the IMJ-PRG, Paris

Homology of a category and the Dehornoy-Lafont order complex, Paris algebra seminar, Paris

2022 **Regular theory in complex braid groups**, Seminario del departamento de Álgebra, Sevilla (Spain).

Théorie régulière dans les groupes de tresses complexes, GATo seminar of the LAMFA, Amiens.

Conferences

2025 **Workshop on trickle groups and their applications**, Tours, from 25/06 to 27/06.

Artin groups, Braids and mapping class groups: Celebrating the work of Luis Paris, Cáceres (Spain), from 16/06 to 20/06 .

Journées jeunes chercheurs et chercheuses en topologie, Amiens, from 09/04 to 11/04.

Colloque tournant du RT algèbre, Clermont-Ferrand, from 03/05 to 03/07.

Winterbraids XIV, Bordeaux, from 02/03 to 02/06.

2023 **2023 Meeting of the GDR Homotopy theory and applications**, Lille, from 10/24 to 10/27.

Various Guises of Reflection Arrangements workshop, Edimbourg (UK), from 03/13 to 03/17.

Spring School on Real, Complex, and Symplectic Reflection Groups, Bochum (Germany), from 03/06 to 03/10.

Winterbraids XII, Tours, from 02/21 to 02/24.

2022 **Journées tresses 2022 : Groupes de tresses généralisés**, Amiens, from 08/29 to 09/02.

2021 **Summer school: The dual approach to Coxeter and Artin groups; Garside theory and applications**, Berlin (Germany), from 08/30 to 09/04.

Teaching

2024–2025 **Exercices courses**, Université de Picardie Jules Verne, Amiens.

- Calculus (first year) (30h).
- Set theory (second year) (20h).
- Groups, Rings, Fields (third year) (36h).
- Linear algebra 1 (first year) (32h).
- parametrized curves (first year) (36h).
- Complex analysis (third year) (20h).

2023–2024 **Exercices courses**, Université de Picardie Jules Verne, Amiens.

- Advanced linear algebra (third year) (20h).
- Complex analysis (third year) (20h).

2022–2023 **Exercices courses**, Université de Picardie Jules Verne, Amiens.

- Advanced linear algebra (third year) (20h).
- Algebraic algorithmics (third year) (20h).

2021–2022 **Exercices courses**, Université de Picardie Jules Verne, Amiens.

- Calculus (first year) (24h).
- Advanced linear algebra (third year) (20h).
- Complex numbers and geometry (third year) (20h).

Other activities

- 2025 **Co-organizer of the workshop on trickle groups and their applications**, Tours.
With Igor Haladjian.
- 2023–2024 **Co-organizer of the PhD student seminar of the LAMFA**, Amiens.
With Nicolas Beuvin et Michaël Schoonheere.
- 2022–2023 **Co-organizer of the PhD student seminar of the LAMFA**, Amiens.
With Felipe Arbulú et Ismaïl Razack.
- 2022 **Research stay in Sevilla**, *Universidad de Sevilla*, Sevilla (Spain).
Work with Juan Gonzalez-Meneses (3 weeks).
- 2022 **Co-organizer of the braid meeting 2022**, Amiens.
With Ivan Marin and Karine Sorlin.